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Stanisław Słowiński

TELECOMMUNICATIONS STUDENT

Motivated engineering student with a strong passion for astronomy and space instrumentation. My academic and project background focuses on hardware-software integration, developing electromechanical prototypes, and designing control systems. I am highly adaptable and eager to apply my hands-on skills in installing, troubleshooting, and maintaining electrical and electronic hardware alongside a team of engineers and technicians

Education

2022 – present
Graduating February 2027

Warsaw University of Technology (WUT)
Bachelor of Engineering in Telecommunications (6th semester)

Latest honors and awards

November 2025

WUT's Headmaster Scholarship for top 8% students based on academic excellence.
A scholarship aimed at the best students based on their performance from two previous semesters.
This is the second time I have received this distinction.

June 2025

2025 IEEE AP-S Student Design Contest finalist
As team leader, I led my group to the finals of the 2025 IEEE AP-S Student Design Contest; we presented our project at the IEEE AP-S/URSI North American Radio Science Meeting in Ottawa, Canada in July 2025.

Personal projects

October 2024 – June 2025

Time Modulated Antenna Array with beamforming capabilities and real-time tracking
Aside from formal duties of a leader (organization, writing reports etc.) I was responsible for many key technical aspects of the project
My roles \ developed skills:

- **Performed hardware assembly and soldering (SMD/THT)**, building a robust physical prototype and utilizing various laboratory equipment for testing.
- **Documented system design and progress by writing detailed reports** and step-by-step instructions to allow reproduction of the system as per the contest's guidelines
- **Programmed microcontrollers and integrated wireless modules (Zigbee)**

- **Designed 3D models** for mechanical components of the setup
- **Designed and implemented a real-time tracking algorithm**, ensuring high positioning accuracy and system reliability.

	September 2025 – present	**Adaptive antenna with beamforming capabilities for tracking weather satellites on the low-earth orbit** Focused on developing a prototype ground station, integrating custom electronics (PCB design in Altium) with software control systems. My roles \ developed skills: • **Designed high frequency circuits in Altium Designer** for a prototype ground station • **Conducted functional testing, troubleshooting and debugging of the electro-mechanical setup** using laboratory equipment to ensure reliable tracking mechanics. • **Integrated and controlled** a satellite tracking rotor using Python within a Linux environment, managing accurate angular positioning.
Student initiatives	March 2025 - present	**Founder and President - Student Club of Antenna Design** I established and I'm currently leading a student club focused on antenna design and applied electromagnetics. Until June 2025 the main focus was the mentioned IEEE contest, however I'm now working on a new project – an antenna array capable of tracking weather satellites on the low Earth orbit (LEO) through the use of electronic beamforming
Community service and volunteering	2023 - present	**Dzieło Na Misji Volunteer** In the past summer of 2024 I taught English and mathematics in a local high school and primary school in rural Tanzania for two months as a volunteer of the Dzieło Na Misji Foundation.
Languages	Polish	Native
	English	C1/C2, full professional proficiency
Computer skills	Programming	Python, Java, MATLAB, C, basics of Verilog
	Software	AutoCAD, Fusion 360, FEKO, Altium Designer, LTspice

Publications

S. Słowiński, M. Bandych, G. Bogdan, and Y. Yashchyshyn, "Sixteen-Element Time-Modulated Antenna Array With Beam Steering in ISM Band," presented at *KRiT*, Gdańsk 2025.